

SAFETY DATA SHEET



XVU91

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	06.01.2026	50002954	Date of first issue: 06.01.2026

SECTION 1. PRODUCT AND COMPANY IDENTIFICATION

Product name : XVU91

Manufacturer or supplier's details

Company : FMC QUÍMICA DO BRASIL LTDA.

Address : AVENIDA DR. JOSÉ BONIFÁCIO
COUTINHO NOGUEIRA 150 - 1º
ANDAR - JARDIM MADALENA,
CAMPINAS SP BRASIL
TELEFONE: (19) 2042.4500

Emergency telephone : Brazil: 0800 34 35 450 (24 hours)
+55-2139581449 (CHEMTREC)

Medical Emergency Number : 0800 7010 450

Recommended use of the chemical and restrictions on use

Recommended use : Insecticide

Restrictions on use : Use as recommended by the label.

SECTION 2. HAZARDS IDENTIFICATION

GHS Classification in accordance with ABNT NBR 14725 Standard

Acute toxicity (Inhalation) : Category 4

Skin corrosion/irritation : Category 3

Serious eye damage/eye irritation : Category 2A

Short-term (acute) aquatic hazard : Category 1

Long-term (chronic) aquatic hazard : Category 1

GHS label elements in accordance with ABNT NBR 14725 Standard

Hazard pictograms :



Signal Word : WARNING

SAFETY DATA SHEET



XVU91

Version 1.0 Revision Date: 06.01.2026 SDS Number: 50002954 Date of last issue: -
Date of first issue: 06.01.2026

Hazard Statements : H316 Causes mild skin irritation.
H319 Causes serious eye irritation.
H332 Harmful if inhaled.
H410 Very toxic to aquatic life with long lasting effects.

Precautionary Statements : **Prevention:**
P261 Avoid breathing dust.
P264 Wash skin thoroughly after handling.
P271 Use only outdoors or in a well-ventilated area.
P273 Avoid release to the environment.
P280 Wear eye protection/ face protection.
Response:
P304 + P340 + P312 IF INHALED: Remove person to fresh air and keep comfortable for breathing. Call a POISON CENTER/ doctor if you feel unwell.
P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P332 + P313 If skin irritation occurs: Get medical advice/ attention.
P337 + P313 If eye irritation persists: Get medical advice/ attention.
P391 Collect spillage.
Disposal:
P501 Dispose of contents/ container to an approved waste disposal plant.

Additional Labeling

The following percentage of the mixture consists of ingredient(s) with unknown hazards to the aquatic environment: 17,75 %

Other hazards which do not result in classification

None known.

SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Mixture

Components

Chemical name	CAS-No.	Classification	Concentration (% w/w)
Chlorantraniliprole	500008-45-7	Aquatic Acute, 1 Aquatic Chronic, 1	>= 25 -< 30
Lignin, alkali, reaction products with formaldehyde and sodium bisulfite	68512-35-6	Serious eye damage/eye irritation, 2A	>= 5 -< 10
kaolin (alternate CAS 1318-74-7)	1332-58-7	Acute Tox. (Inhalation), 5	>= 5 -< 10
citric acid	77-92-9	Serious eye damage/eye irritation, 2A	>= 1 -< 5

SAFETY DATA SHEET



XVU91

Version 1.0 Revision Date: 06.01.2026 SDS Number: 50002954 Date of last issue: -
Date of first issue: 06.01.2026

		STOT SE, (Respiratory system) , 3	
sodium dimethylnaphthalenesulphonate	27178-87-6	Acute Tox. (Oral), 5 Skin corrosion/irritation, 2 Serious eye damage/eye irritation, 1 Aquatic Acute, 3	$\geq 2,5 - < 3$
Alkylated Naphthalene Sulfonate Sodium Salt	68425-94-5	Serious eye damage/eye irritation, 2A Aquatic Acute, 3 Aquatic Chronic, 3	$\geq 1 - < 2,5$

SECTION 4. FIRST AID MEASURES

- General advice : Move out of dangerous area.
Show this material safety data sheet to the doctor in attendance.
Do not leave the victim unattended.
- If inhaled : Consult a physician after significant exposure.
If unconscious, place in recovery position and seek medical advice.
- In case of skin contact : If on clothes, remove clothes.
Wash off with soap and water.
If symptoms persist, call a physician.
Wash contaminated clothing before re-use.
- In case of eye contact : Immediately flush eye(s) with plenty of water.
Remove contact lenses.
Protect unharmed eye.
Keep eye wide open while rinsing.
If eye irritation persists, consult a specialist.
- If swallowed : Keep respiratory tract clear.
Do not give milk or alcoholic beverages.
Never give anything by mouth to an unconscious person.
If symptoms persist, call a physician.
- Most important symptoms and effects, both acute and delayed : Causes mild skin irritation.
Causes serious eye irritation.
Harmful if inhaled.
- Protection of first-aiders : Avoid inhalation, ingestion and contact with skin and eyes.
- Notes to physician : Treat symptomatically.

SECTION 5. FIRE-FIGHTING MEASURES

SAFETY DATA SHEET



XVU91

Version 1.0	Revision Date: 06.01.2026	SDS Number: 50002954	Date of last issue: - Date of first issue: 06.01.2026
----------------	------------------------------	-------------------------	--

- | | |
|--|--|
| Suitable extinguishing media | : Dry chemical, CO2, water spray or regular foam. |
| Unsuitable extinguishing media | : Do not spread spilled material with high-pressure water streams. |
| Specific hazards during fire fighting | : Do not allow run-off from fire fighting to enter drains or water courses. |
| Hazardous combustion products | : Fire may produce irritating, corrosive and/or toxic gases.
Nitrogen oxides (NOx)
Carbon oxides
Bromine compounds
Hydrogen cyanide
Hydrogen chloride
Chlorinated compounds
phosphorus oxides |
| Specific extinguishing methods | : Use a water spray to cool fully closed containers.
Remove undamaged containers from fire area if it is safe to do so.
Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Collect contaminated fire extinguishing water separately. This must not be discharged into drains.
Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. |
| Special protective equipment for fire-fighters | : Firefighters should wear protective clothing and self-contained breathing apparatus. |

SECTION 6. ACCIDENTAL RELEASE MEASURES

- | | |
|---|---|
| Personal precautions, protective equipment and emergency procedures | : If it can be safely done, stop the leak.
Do not touch or walk through the spilled material.
Use personal protective equipment.
Evacuate personnel to safe areas.
Avoid dust formation.
Avoid breathing dust.
Ensure adequate ventilation. |
| Environmental precautions | : Prevent product from entering drains.
Prevent further leakage or spillage if safe to do so.
If the product contaminates rivers and lakes or drains inform respective authorities. |
| Methods and materials for containment and cleaning up | : Never return spills in original containers for re-use. Pick up and transfer the spilled material to a properly labeled container without creating dust. For spills on concrete or other non-porous surfaces, the area can be cleaned using a small quantity of soap and water. Do not allow the cleaning solution to enter drains. Use an inert absorbent material to soak up the cleaning solution and transfer it to the properly labeled container. When the spill occurs on soil, the only effective way to |

SAFETY DATA SHEET



XVU91

Version 1.0 Revision Date: 06.01.2026 SDS Number: 50002954 Date of last issue: -
Date of first issue: 06.01.2026

decontaminate the area is to remove the top 5 to 7 centimeters of soil.

SECTION 7. HANDLING AND STORAGE

- Advice on protection against fire and explosion : Avoid dust formation.
Provide appropriate exhaust ventilation at places where dust is formed.
- Advice on safe handling : Avoid formation of respirable particles.
Do not breathe vapors/dust.
Avoid contact with skin and eyes.
For personal protection see section 8.
Smoking, eating and drinking should be prohibited in the application area.
Provide sufficient air exchange and/or exhaust in work rooms.
Dispose of rinse water in accordance with local and national regulations.
- Hygiene measures : Avoid contact with skin, eyes and clothing.
Do not breathe dust.
When using do not eat or drink.
When using do not smoke.
Wash hands before breaks and at the end of workday.
- Conditions for safe storage : Keep container tightly closed in a dry and well-ventilated place.
Containers which are opened must be carefully resealed and kept upright to prevent leakage.
Electrical installations / working materials must comply with the technological safety standards.
- Further information on storage stability : No decomposition if stored and applied as directed.

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Ingredients with workplace control parameters

Components	CAS-No.	Value type (Form of exposure)	Control parameters / Permissible concentration	Basis
kaolin	1332-58-7	LT	8,5 mppcd / (% quartz+10) (Silica)	BR OEL
		LT (Respirable dust)	8 mg/m3 / (% quartz+2) (Silica)	BR OEL
		LT (Total dust)	24 mg/m3 / (% quartz+3) (Silica)	BR OEL
		TWA (Respirable par-	2 mg/m3	ACGIH

SAFETY DATA SHEET



XVU91

Version 1.0	Revision Date: 06.01.2026	SDS Number: 50002954	Date of last issue: - Date of first issue: 06.01.2026
----------------	------------------------------	-------------------------	--

		ticulate mat- ter)		
--	--	-----------------------	--	--

Personal protective equipment

Respiratory protection	:	In the case of dust or aerosol formation use respirator with an approved filter.
Filter type	:	Particulates type
Hand protection	:	
Material	:	Protective gloves
Remarks	:	The suitability for a specific workplace should be discussed with the producers of the protective gloves.
Eye protection	:	Eye wash bottle with pure water Tightly fitting safety goggles
Skin and body protection	:	Dust impervious protective suit Choose body protection according to the amount and concentration of the dangerous substance at the work place.
Protective measures	:	Plan first aid action before beginning work with this product.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state	:	solid
Form	:	granules
Color	:	No data available
Odor	:	No data available
Odor Threshold	:	No data available
pH	:	No data available
Melting point/ range	:	No data available
Boiling point/boiling range	:	No data available
Flash point	:	Not applicable
Evaporation rate	:	Not applicable
Flammability (solid, gas)	:	The product may be combustible. Method: US EPA Method 1030 (Ignitability of Solids)

SAFETY DATA SHEET



XVU91

Version 1.0	Revision Date: 06.01.2026	SDS Number: 50002954	Date of last issue: - Date of first issue: 06.01.2026
----------------	------------------------------	-------------------------	--

Self-ignition	:	No data available
Upper explosion limit / Upper flammability limit	:	No data available
Lower explosion limit / Lower flammability limit	:	No data available
Relative vapor density	:	Not applicable
Relative density	:	No data available
Density	:	No data available
Solubility(ies) Water solubility	:	Miscible (20 - 30 °C)
Partition coefficient: n-octanol/water	:	No data available
Autoignition temperature	:	No data available
Decomposition temperature	:	No data available
Viscosity Viscosity, dynamic	:	Not applicable
Viscosity, kinematic	:	Not applicable
Explosive properties	:	Not explosive
Oxidizing properties	:	Non-oxidizing
Surface tension	:	52,3 mN/m, OECD Test Guideline 115
Molecular weight	:	Not applicable
Metal corrosion rate	:	Not corrosive to metals.
Particle size	:	No data available

SECTION 10. STABILITY AND REACTIVITY

Reactivity	:	No decomposition if stored and applied as directed.
Chemical stability	:	No decomposition if stored and applied as directed.
Possibility of hazardous reactions	:	No decomposition if stored and applied as directed.

SAFETY DATA SHEET



XVU91

Version 1.0	Revision Date: 06.01.2026	SDS Number: 50002954	Date of last issue: - Date of first issue: 06.01.2026
----------------	------------------------------	-------------------------	--

Conditions to avoid	: Avoid extreme temperatures. Avoid dust formation.
Incompatible materials	: Avoid strong acids, bases, and oxidizers.
Hazardous decomposition products	: No hazardous decomposition products are known.

SECTION 11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation
Skin contact

Acute toxicity

Harmful if inhaled.

Product:

Acute oral toxicity : LD50 (Rat, female): > 5.000 mg/kg
Method: OECD Test Guideline 425
GLP: yes
Assessment: The substance or mixture has no acute oral toxicity
Remarks: no mortality

Acute inhalation toxicity : LC50 (Rat, male and female): 1,03 - 5,33 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Method: OECD Test Guideline 403
Symptoms: Breathing difficulties, Fatality
GLP: yes

Acute dermal toxicity : LD50 (Rat, male and female): > 5.000 mg/kg
Method: OECD Test Guideline 402
GLP: yes
Assessment: The substance or mixture has no acute dermal toxicity
Remarks: no mortality

Components:

Chlorantraniliprole:

Acute oral toxicity : LD50 (Rat, female): > 5.000 mg/kg
Method: OECD Test Guideline 425
GLP: yes
Assessment: The substance or mixture has no acute oral toxicity

LD50 (Rat): > 5.000 mg/kg
Method: OECD Test Guideline 425
GLP: yes
Remarks: Information source: Internal study report

XVU91

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	06.01.2026	50002954	Date of first issue: 06.01.2026

LD50 (Mouse, female): > 2.000 mg/kg
Method: OECD Test Guideline 425
GLP: no

Acute inhalation toxicity : LC50 (Rat, male and female): > 5,1 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Method: OECD Test Guideline 403
GLP: yes
Assessment: The substance or mixture has no acute inhalation toxicity
Remarks: Information source: Internal study report

LC50 (Rat, male and female): > 5,1 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Method: OECD Test Guideline 403
GLP: yes
Assessment: The substance or mixture has no acute inhalation toxicity
Remarks: no mortality

LC50 (Rat, male and female): > 5,0 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Method: GB 15670-1995
GLP: yes
Assessment: The substance or mixture has no acute inhalation toxicity
Remarks: no mortality

Acute dermal toxicity : LD50 (Rat, male and female): > 5.000 mg/kg
Method: OECD Test Guideline 402
GLP: yes
Assessment: The substance or mixture has no acute dermal toxicity
Remarks: Information source: Internal study report

LD50 (Rat, male and female): > 5.000 mg/kg
Method: GB 15670-1995
GLP: yes
Remarks: no mortality

LD50 (Rat, male and female): > 5.000 mg/kg
Method: OECD Test Guideline 402
GLP: yes
Remarks: no mortality

Lignin, alkali, reaction products with formaldehyde and sodium bisulfite:

Acute oral toxicity : LD50 (Rat, male and female): > 5.000 mg/kg

kaolin:

Acute oral toxicity : LD50 (Rat): > 5.000 mg/kg

SAFETY DATA SHEET



XVU91

Version 1.0	Revision Date: 06.01.2026	SDS Number: 50002954	Date of last issue: - Date of first issue: 06.01.2026
----------------	------------------------------	-------------------------	--

Method: OECD Test Guideline 401

LD50: > 2.000 mg/kg

Method: OECD Test Guideline 420

Assessment: The substance or mixture has no acute oral toxicity

Acute inhalation toxicity : LC50: 5,07 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Method: OECD Test Guideline 436

Acute dermal toxicity : LD50 (Rat): > 5.000 mg/kg

LD50: > 2.000 mg/kg

Method: OECD Test Guideline 402

Assessment: The substance or mixture has no acute dermal toxicity

citric acid:

Acute oral toxicity : LD50 (Mouse, male and female): 5.400 mg/kg
Method: OECD Test Guideline 401

Acute dermal toxicity : LD50 (Rat, male and female): > 2.000 mg/kg
Method: OECD Test Guideline 402
Assessment: The substance or mixture has no acute dermal toxicity

sodium dimethylnaphthalenesulphonate:

Acute oral toxicity : LD50 (Rat): > 2.000 - 5.000 mg/kg
Method: OECD Test Guideline 401

LD50 (Rat): > 3.000 - 5.000 mg/kg

Method: OECD Test Guideline 401

Remarks: Based on data from similar materials

Acute dermal toxicity : LD50 (Rabbit): > 5.000 mg/kg
Method: OECD Test Guideline 404
Remarks: Based on data from similar materials

Alkylated Naphthalene Sulfonate Sodium Salt:

Acute oral toxicity : LD50 (Rat): > 5.000 mg/kg

Skin corrosion/irritation

Causes mild skin irritation.

Product:

Species : Rabbit
Assessment : Not classified as irritant
Method : OECD Test Guideline 404
Result : slight irritation
GLP : yes

SAFETY DATA SHEET



XVU91

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	06.01.2026	50002954	Date of first issue: 06.01.2026

Remarks : May cause skin irritation and/or dermatitis.

Components:

Chlorantraniliprole:

Species : Rabbit
Method : OECD Test Guideline 404
Result : No skin irritation
GLP : yes
Remarks : Information source: Internal study report

Species : Rabbit
Method : OECD Test Guideline 404
Result : No skin irritation
GLP : yes

Species : Rabbit
Method : GB 15670-1995
Result : No skin irritation
GLP : yes

Lignin, alkali, reaction products with formaldehyde and sodium bisulfite:

Result : No skin irritation

kaolin:

Method : OECD Test Guideline 404
Result : No skin irritation

citric acid:

Species : Rabbit
Method : OECD Test Guideline 404
Result : No skin irritation

sodium dimethylnaphthalenesulphonate:

Species : Rabbit
Method : OECD Test Guideline 404
Result : Skin irritation

Alkylated Naphthalene Sulfonate Sodium Salt:

Remarks : No data available

Serious eye damage/eye irritation

Causes serious eye irritation.

Product:

Species : Rabbit
Result : slight irritation
Assessment : Not classified as irritant
Method : OECD Test Guideline 405

SAFETY DATA SHEET



XVU91

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	06.01.2026	50002954	Date of first issue: 06.01.2026

GLP : yes

Remarks : May cause irreversible eye damage.

Components:

Chlorantraniliprole:

Species : Rabbit

Result : No eye irritation

Method : OECD Test Guideline 405

GLP : yes

Remarks : Information source: Internal study report

Species : Rabbit

Result : No eye irritation

Method : OECD Test Guideline 405

Species : Rabbit

Result : Slight or no eye irritation

Assessment : Not classified as irritant

Method : OECD Test Guideline 405

GLP : yes

Lignin, alkali, reaction products with formaldehyde and sodium bisulfite:

Result : Moderate eye irritation

kaolin:

Result : No eye irritation

Method : OECD Test Guideline 405

citric acid:

Species : Rabbit

Result : Irritation to eyes, reversing within 21 days

Method : OECD Test Guideline 405

sodium dimethylnaphthalenesulphonate:

Result : Irreversible effects on the eye

Method : OECD Test Guideline 437

Species : Rabbit

Result : Irreversible effects on the eye

Method : OECD Test Guideline 405

Remarks : Based on data from similar materials

Alkylated Naphthalene Sulfonate Sodium Salt:

Result : Eye irritation

XVU91

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	06.01.2026	50002954	Date of first issue: 06.01.2026

Respiratory or skin sensitization**Skin sensitization**

Based on available data, the classification criteria are not met.

Respiratory sensitization

Based on available data, the classification criteria are not met.

Product:

Test Type	: Local lymph node assay (LLNA)
Routes of exposure	: Dermal
Species	: Mouse
Assessment	: Not a skin sensitizer.
Method	: OECD Test Guideline 429
GLP	: yes

Components:**Chlorantraniliprole:**

Test Type	: Maximization Test
Species	: Guinea pig
Method	: OECD Test Guideline 406
Result	: Does not cause skin sensitization.
GLP	: yes
Remarks	: Information source: Internal study report

Test Type	: Local lymph node assay (LLNA)
Species	: mice
Method	: OECD Test Guideline 429
Result	: Does not cause skin sensitization.

Lignin, alkali, reaction products with formaldehyde and sodium bisulfite:

Species	: Guinea pig
Result	: Not a skin sensitizer.

kaolin:

Method	: OECD Test Guideline 429
Result	: Does not cause skin sensitization.

sodium dimethylnaphthalenesulphonate:

Result	: Does not cause skin sensitization.
--------	--------------------------------------

Germ cell mutagenicity

Based on available data, the classification criteria are not met.

Product:

Genotoxicity in vitro	: Test Type: Ames test Metabolic activation: with and without metabolic activation Method: OECD Test Guideline 471 Result: negative GLP: yes Remarks: Based on data from similar materials
-----------------------	---

SAFETY DATA SHEET



XVU91

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	06.01.2026	50002954	Date of first issue: 06.01.2026

Test Type: Micronucleus test
Test system: Human lymphocytes
Method: OECD Test Guideline 487
Result: negative
Remarks: Based on data from similar materials

Genotoxicity in vivo : Test Type: Micronucleus test

Components:

Chlorantraniliprole:

Genotoxicity in vitro : Test Type: reverse mutation assay
Metabolic activation: with and without metabolic activation
Result: negative

Test Type: In vitro mammalian cell gene mutation test
Test system: Chinese hamster ovary cells
Method: OECD Test Guideline 476
Result: negative

Genotoxicity in vivo : Test Type: Micronucleus test
Species: Mouse
Method: OECD Test Guideline 474
Result: negative

Germ cell mutagenicity - Assessment : Weight of evidence does not support classification as a germ cell mutagen.

Lignin, alkali, reaction products with formaldehyde and sodium bisulfite:

Genotoxicity in vitro : Test Type: reverse mutation assay
Method: OECD Test Guideline 471
Result: negative

kaolin:

Genotoxicity in vitro : Test Type: Ames test
Method: OECD Test Guideline 471
Result: negative

Genotoxicity in vivo : Remarks: No data available

citric acid:

Genotoxicity in vitro : Test Type: reverse mutation assay
Result: negative

Genotoxicity in vivo : Test Type: Cytogenetic assay
Species: Rat
Application Route: Oral
Result: negative

sodium dimethylnaphthalenesulphonate:

SAFETY DATA SHEET



XVU91

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	06.01.2026	50002954	Date of first issue: 06.01.2026

Genotoxicity in vitro : Method: OECD Test Guideline 471
Result: negative

Method: OECD Test Guideline 476
Result: negative

Carcinogenicity

Based on available data, the classification criteria are not met.

Components:

Chlorantraniliprole:

Species : Rat, male and female
Application Route : Oral
Exposure time : 2 Years
NOAEL : 805 - 1.076 mg/kg bw/day
Method : OECD Test Guideline 453
Result : negative

Species : Mouse, male and female
Application Route : Oral
Exposure time : 18 month(s)
NOAEL : 158 - 1.155 mg/kg bw/day
Method : OECD Test Guideline 453
Result : negative

Species : Dog
Exposure time : 1 Years
NOAEL : 1.164 mg/kg bw/day
Method : OECD Test Guideline 453
Result : negative

Carcinogenicity - Assessment : Weight of evidence does not support classification as a carcinogen

Reproductive toxicity

Based on available data, the classification criteria are not met.

Components:

Chlorantraniliprole:

Effects on fertility : Test Type: Two-generation study
Species: Rat, male and female
Application Route: Oral
General Toxicity Parent: NOAEL: 20.000 ppm
General Toxicity F1: NOAEL: 20.000 ppm
Method: OECD Test Guideline 416
Result: negative

Effects on fetal development : Test Type: Pre-natal
Species: Rat
Application Route: Oral
Duration of Single Treatment: 6 - 20 Days
General Toxicity Maternal: NOEL: 1.000 mg/kg bw/day

SAFETY DATA SHEET



XVU91

Version 1.0	Revision Date: 06.01.2026	SDS Number: 50002954	Date of last issue: - Date of first issue: 06.01.2026
----------------	------------------------------	-------------------------	--

Developmental Toxicity: NOEL: 1.000 mg/kg bw/day
Method: OECD Test Guideline 414
Result: negative

Reproductive toxicity - Assessment : Weight of evidence does not support classification for reproductive toxicity

kaolin:

Effects on fertility : Remarks: No data available

Effects on fetal development : Remarks: No data available

STOT-single exposure

Based on available data, the classification criteria are not met.

Components:

kaolin:

Remarks : No significant adverse effects were reported

citric acid:

Assessment : May cause respiratory irritation.

STOT-repeated exposure

Based on available data, the classification criteria are not met.

Components:

kaolin:

Assessment : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

Repeated dose toxicity

Components:

Chlorantraniliprole:

Species	: Rat, male and female
NOEL	: 1188 - 1526 mg/kg
Application Route	: Oral
Exposure time	: 90 Days
Method	: OECD Test Guideline 408

kaolin:

Remarks : No data available

Aspiration toxicity

Based on available data, the classification criteria are not met.

XVU91

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	06.01.2026	50002954	Date of first issue: 06.01.2026

Components:**Chlorantraniliprole:**

The substance does not have properties associated with aspiration hazard potential.

Further information**Product:**

Remarks : No data available

SECTION 12. ECOLOGICAL INFORMATION**Ecotoxicity****Product:**

- | | | |
|---|---|--|
| Toxicity to fish | : | LC50 (Oncorhynchus mykiss (rainbow trout)): > 9,4 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203
GLP: yes
Remarks: Based on data from a similar product. |
| Toxicity to daphnia and other aquatic invertebrates | : | EC50 (Daphnia magna (Water flea)): 0,038 mg/l
End point: Immobilization
Exposure time: 48 h
Method: OECD Test Guideline 202
GLP: yes
Remarks: Based on data from a similar product. |
| Toxicity to algae/aquatic plants | : | EC50 (Raphidocelis subcapitata (freshwater green alga)): > 20,2 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201
Remarks: Based on data from a similar product. |
| Toxicity to soil dwelling organisms | : | Method: OECD Test Guideline 216
GLP: yes
Remarks: No significant adverse effect on Nitrogen mineralization.

Method: OECD Test Guideline 217
GLP: yes
Remarks: No significant adverse effect on Carbon mineralization.

NOEC (Eisenia fetida (earthworms)): > 1.000 mg/kg
Method: OECD Test Guideline 207
GLP: yes |
| Toxicity to terrestrial organisms | : | LD50 (Apis mellifera (bees)): > 200
Exposure time: 48 h
Method: OECD Test Guideline 214
GLP: yes
Remarks: Contact |

XVU91

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	06.01.2026	50002954	Date of first issue: 06.01.2026

LD50 (*Apis mellifera* (bees)): > 191
Exposure time: 48 h
Method: OECD Test Guideline 213
GLP: yes
Remarks: Oral

LD50 (*Colinus virginianus* (Bobwhite quail)): > 2000
Exposure time: 14 d
Method: OECD Test Guideline 223
GLP: yes

Components:**Chlorantraniliprole:**

Toxicity to fish : LC50 (*Oncorhynchus mykiss* (rainbow trout)): 13,8 mg/l
Exposure time: 96 h
Test Type: static test
Method: OECD Test Guideline 203
Remarks: Information source: Internal study report

LC50 (*Lepomis macrochirus* (Bluegill sunfish)): > 15,1 mg/l
Exposure time: 96 h
Test Type: static test
Method: OECD Test Guideline 203
GLP: yes
Remarks: Information source: Internal study report

LC50 (*Cyprinodon* sp. (minnow)): > 12 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203

Toxicity to daphnia and other aquatic invertebrates : EC50 (*Daphnia magna* (Water flea)): 0,0116 mg/l
Exposure time: 48 h
Test Type: static test
Method: OECD Test Guideline 202
GLP: yes

LC50 (*Hyalella azteca* (Amphipod)): 0,26 mg/l
Exposure time: 48 h
Test Type: static test
Method: OECD Test Guideline 202
GLP: yes

LC50 (*Ceriodaphnia dubia* (water flea)): 0,0067 - 0,011 mg/l
Exposure time: 48 h

Toxicity to algae/aquatic plants : ErC50 (*Pseudokirchneriella subcapitata* (green algae)): > 2 mg/l
Exposure time: 120 h

NOEC (*Lemna gibba* (duckweed)): > 2 mg/l
End point: Biomass
Exposure time: 14 d
Test Type: static test

SAFETY DATA SHEET



XVU91

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	06.01.2026	50002954	Date of first issue: 06.01.2026

ErC50 (*Selenastrum capricornutum* (green algae)): > 2 mg/l
Exposure time: 72 h

NOEC (*Anabaena flos-aquae* (cyanobacterium)): > 2 mg/l
End point: Growth rate
Exposure time: 120 h
Test Type: static test
Method: OECD Test Guideline 201
GLP: yes

NOEC (*Skeletonema costatum* (Diatom)): > 14,6 mg/l
End point: Growth rate
Exposure time: 120 h
Test Type: static test
Method: OECD Test Guideline 201
GLP: yes

NOEC (*Navicula pelliculosa* (Diatom)): > 15,1 mg/l
End point: Growth rate
Exposure time: 120 h
Test Type: static test
Method: OECD Test Guideline 201
GLP: yes

M-Factor (Acute aquatic toxicity) : 10

Toxicity to fish (Chronic toxicity) : NOEC (*Cyprinodon variegatus* (sheepshead minnow)): 1,28 mg/l
Exposure time: 36 d

NOEC (*Oncorhynchus mykiss* (rainbow trout)): 0,110 mg/l
Exposure time: 28 d
Method: OECD Test Guideline 210
GLP: yes

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC (*Daphnia magna* (Water flea)): 0,00447 mg/l
Exposure time: 21 d
Method: US EPA Test Guideline OPPTS 850.1300
GLP: yes

M-Factor (Chronic aquatic toxicity) : 10

Toxicity to soil dwelling organisms : LC50 (*Eisenia fetida* (earthworms)): > 1.000 mg/kg
Exposure time: 14 d
Method: OECD Test Guideline 207
GLP: yes

Method: OECD Test Guideline 216
Remarks: No significant adverse effect on Nitrogen mineralization.

Method: OECD Test Guideline 217

XVU91

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	06.01.2026	50002954	Date of first issue: 06.01.2026

Remarks: No significant adverse effect on Carbon mineralization.

EC50 (*Hypoaspis aculeifer*): >100 mg/kg dry weight (d.w.)
Exposure time: 16 d
Method: OECD Test Guideline 207

NOEC (*Hypoaspis aculeifer*): 100 mg/kg dry weight (d.w.)
Exposure time: 16 d
Method: OECD Test Guideline 207

Toxicity to terrestrial organisms : LD50 (*Apis mellifera* (bees)): > 4,0 µg/bee
Exposure time: 72 h
End point: Acute contact toxicity
Remarks: Active substance dissolved in acetone

LD50 (*Apis mellifera* (bees)): > 0,005 µg/bee
Exposure time: 48 h
End point: Acute contact toxicity
Remarks: Active substance dissolved in water

LD50 (*Apis mellifera* (bees)): > 104,1 µg/bee
Exposure time: 48 h
End point: Acute oral toxicity
Remarks: Active substance dissolved in acetone

LD50 (*Apis mellifera* (bees)): > 0,0274 µg/bee
Exposure time: 48 h
End point: Acute oral toxicity
Remarks: Active substance dissolved in water

LD50 (*Poephila guttata* (zebra finch)): > 2.250 mg/kg

Lignin, alkali, reaction products with formaldehyde and sodium bisulfite:

Toxicity to fish : LC50 (*Pimephales promelas* (fathead minnow)): 615 mg/l
Exposure time: 96 h

kaolin:

Toxicity to fish : LC50 (*Oncorhynchus mykiss* (rainbow trout)): > 100 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203

Toxicity to daphnia and other aquatic invertebrates : EC50 (*Daphnia magna* (Water flea)): > 1.000 mg/l
Exposure time: 48 h
Method: OECD Test Guideline 202

Toxicity to algae/aquatic plants : EC50 (*Raphidocelis subcapitata* (freshwater green alga)): > 100 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : Remarks: No data available

XVU91

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	06.01.2026	50002954	Date of first issue: 06.01.2026

Toxicity to microorganisms : Remarks: No data available

citric acid:

Toxicity to fish : LC50 (Pimephales promelas (fathead minnow)): > 100 mg/l
Exposure time: 96 h

Toxicity to daphnia and other : EC50 (Daphnia magna (Water flea)): 1.535 mg/l
aquatic invertebrates Exposure time: 24 h

sodium dimethylnaphthalenesulphonate:

Toxicity to fish : LC50 (Danio rerio (zebra fish)): > 10 - 100 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203

Toxicity to daphnia and other : EC50 (Daphnia magna (Water flea)): > 100 mg/l
aquatic invertebrates Exposure time: 48 h
Method: OECD Test Guideline 202

Toxicity to algae/aquatic : EC10 (Pseudokirchneriella subcapitata (green algae)): 135
plants mg/l
Exposure time: 72 h
Test Type: static test
Method: OECD Test Guideline 201

EC50 (Pseudokirchneriella subcapitata (green algae)): > 810
mg/l
Exposure time: 72 h
Test Type: static test
Method: OECD Test Guideline 201

Toxicity to daphnia and other : EC10 (Daphnia magna (Water flea)): > 1 - 10 mg/l
aquatic invertebrates (Chron- Exposure time: 21 d
ic toxicity) Method: OECD Test Guideline 211

Toxicity to microorganisms : EC10 (Pseudomonas putida): > 100 mg/l
Exposure time: 16,5 h
Method: DIN 38 412 Part 8
Remarks: Based on data from similar materials

Ecotoxicology Assessment

Chronic aquatic toxicity : This product has no known ecotoxicological effects.

Alkylated Naphthalene Sulfonate Sodium Salt:

Toxicity to fish : LC50 (Zebra fish): > 10 - 100 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203
Remarks: Based on data from similar materials

Toxicity to daphnia and other : EC50 (Daphnia magna (Water flea)): > 100 mg/l
aquatic invertebrates Exposure time: 48 h
Method: OECD Test Guideline 202

XVU91

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	06.01.2026	50002954	Date of first issue: 06.01.2026

Remarks: Based on data from similar materials

Toxicity to algae/aquatic plants : EC50 (Pseudokirchneriella subcapitata (green algae)): > 100 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201
Remarks: Based on data from similar materials

EC10 (Pseudokirchneriella subcapitata (green algae)): > 100 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201
Remarks: Based on data from similar materials

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : EC10 (Daphnia magna (Water flea)): > 10 - 100 mg/l
Exposure time: 21 d
Method: OECD Test Guideline 211
Remarks: Based on data from similar materials

Persistence and degradability**Components:****Chlorantraniliprole:**

Biodegradability : Result: Not readily biodegradable.

Stability in water : Degradation half life (DT50): 10 d (25 °C) pH: 9
Degradation half life (DT50): 0,3 d (50 °C) pH: 9
Degradation half life (DT50): > 31 d pH: 5

Lignin, alkali, reaction products with formaldehyde and sodium bisulfite:

Biodegradability : Result: Not readily biodegradable.
Biodegradation: < 5 %
Exposure time: 28 d
Method: OECD Test Guideline 301E

kaolin:

Biodegradability : Remarks: The methods for determining biodegradability are not applicable to inorganic substances.

citric acid:

Biodegradability : Result: Readily biodegradable.
Biodegradation: 97 %
Exposure time: 28 d
Method: OECD Test Guideline 301B

sodium dimethylnaphthalenesulphonate:

Biodegradability : Result: Inherently biodegradable.
Method: OECD Test Guideline 301D

SAFETY DATA SHEET



XVU91

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	06.01.2026	50002954	Date of first issue: 06.01.2026

Alkylated Naphthalene Sulfonate Sodium Salt:

Biodegradability : Result: Not readily biodegradable.
Remarks: Based on data from similar materials

Bioaccumulative potential

Components:

Chlorantraniliprole:

Bioaccumulation : Species: Lepomis macrochirus (Bluegill sunfish)
Bioconcentration factor (BCF): 14
Method: OECD Test Guideline 305
GLP: yes
Remarks: Bioaccumulation is unlikely.

Partition coefficient: n-octanol/water : log Pow: 2,77 (20 °C)
pH: 4

log Pow: 2,86 (20 °C)
pH: 7

log Pow: 2,80 (20 °C)
pH: 9

kaolin:

Bioaccumulation : Remarks: Bioaccumulation is unlikely.

Partition coefficient: n-octanol/water : Remarks: Not applicable

Mobility in soil

Components:

Chlorantraniliprole:

Distribution among environmental compartments : Koc: 362 ml/g, log Koc: 2,55
Remarks: Mobile in soils

Stability in soil : Remarks: Very persistent in soil.

kaolin:

Distribution among environmental compartments : Remarks: Low mobility in soil.

Other adverse effects

Product:

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.
Very toxic to aquatic life with long lasting effects.

XVU91

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	06.01.2026	50002954	Date of first issue: 06.01.2026

SECTION 13. DISPOSAL CONSIDERATIONS**Disposal methods**

Waste from residues : The product should not be allowed to enter drains, water courses or the soil.
Do not contaminate ponds, waterways or ditches with chemical or used container.
Send to a licensed waste management company.

Contaminated packaging : It is prohibited to reuse, bury, burn or sell packaging.

Washable packaging: Triple wash packs of less than 20 liters and pressure wash packs of 20 liters or more. Triple Wash (Manual Wash): Completely empty the contents of the package into the sprayer tank, keeping it in an upright position for 30 seconds; Add clean water to the package up to $\frac{1}{4}$ of its volume; Cover the package well and shake it for 30 seconds; Pour the wash water into the spray tank; Do this operation three times; Make the plastic or metal packaging unusable by perforating the bottom.

Pressure wash: Fit the empty package in the appropriate place of the funnel installed on the sprayer; Activate the mechanism to release the water jet; Direct the water jet to all the inside walls of the package, for 30 seconds; Wash water must be transferred to the sprayer tank; Make the plastic or metal packaging unusable by perforating the bottom. In both procedures, puncture the container at its base without damaging the label. Within a period of up to one year from the date of purchase, the user must return the empty packaging, with lid, to the establishment where the product was purchased or to the place indicated on the invoice, issued at the time of purchase. Activate the mechanism to release the water jet. Direct the water jet to all the inside walls of the package, for 30 seconds. Wash water must be transferred to the sprayer tank. Make the plastic or metal packaging unusable by perforating the bottom.

SECTION 14. TRANSPORT INFORMATION**International Regulations****UNRTDG**

UN number	:	UN 3077
Proper shipping name	:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (Chlorantraniliprole)
Class	:	9
Subsidiary risk	:	ENVIRONM.
Packing group	:	III
Labels	:	9 (ENVIRONM.)
Environmentally hazardous	:	no

SAFETY DATA SHEET



XVU91

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	06.01.2026	50002954	Date of first issue: 06.01.2026

IATA-DGR

UN/ID No.	: UN 3077
Proper shipping name	: Environmentally hazardous substance, solid, n.o.s. (Chlorantraniliprole)
Class	: 9
Packing group	: III
Labels	: Miscellaneous
Packing instruction (cargo aircraft)	: 956
Packing instruction (passenger aircraft)	: 956

IMDG-Code

UN number	: UN 3077
Proper shipping name	: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (Chlorantraniliprole)
Class	: 9
Packing group	: III
Labels	: 9
EmS Code	: F-A, S-F
Marine pollutant	: yes

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

Domestic regulation

ANTT

UN number	: UN 3077
Proper shipping name	: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (Chlorantraniliprole)
Class	: 9
Packing group	: III
Labels	: 9
Hazard Identification Number	: 90

Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

SECTION 15. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

Law No. 14,785 of December 27, 2023. Decree 4,074 of January 4, 2002 and its regulatory standards. ANTT Resolution No. 5,998/22 of November 3, 2022. This SDS was prepared in accordance with the criteria of ABNT NBR 14725. The user is recommended to pay attention to local regulations.

Law No. 14,785 of December 27, 2023. Decree 4,074 of January 4, 2002 and its regulatory standards. ANTT Resolution No. 5,998/22 of November 3, 2022. This SDS was prepared in accordance with the criteria of ABNT NBR 14725. The user is recommended to pay attention to local regulations.

SAFETY DATA SHEET



XVU91

Version 1.0	Revision Date: 06.01.2026	SDS Number: 50002954	Date of last issue: - Date of first issue: 06.01.2026
----------------	------------------------------	-------------------------	--

National List of Carcinogenic Agents for Humans - (LINACH)

Group 1: Carcinogenic to humans
kaolin

1332-58-7

Brazil. List of chemicals controlled by the Federal Police : Not applicable

The ingredients of this product are reported in the following inventories:

TCSI	: Not in compliance with the inventory
TSCA	: Product contains substance(s) not listed on TSCA inventory.
AIIC	: Not in compliance with the inventory
DSL	: This product contains the following components that are not on the Canadian DSL nor NDSL. Chlorantraniliprole Bacillus thuringiensis Lignin, alkali, reaction products with formaldehyde and sodium bisulfite
ENCS	: Not in compliance with the inventory
ISHL	: Not in compliance with the inventory
KECI	: Not in compliance with the inventory
PICCS	: Not in compliance with the inventory
IECSC	: Not in compliance with the inventory
NZIoC	: Not in compliance with the inventory
TECI	: Not in compliance with the inventory

SECTION 16. OTHER INFORMATION

Revision Date	: 06.01.2026
Date format	: dd.mm.yyyy

Full text of other abbreviations

ACGIH	: USA. ACGIH Threshold Limit Values (TLV)
BR OEL	: Brazil. NR 15 - Unhealthy activities and operations
ACGIH / TWA	: 8-hour, time-weighted average
BR OEL / LT	: Up to 48 hours /week

XVU91

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	06.01.2026	50002954	Date of first issue: 06.01.2026

AIIC - Australian Inventory of Industrial Chemicals; ANTT - National Agency for Transport by Land of Brazil; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; Nch - Chilean Norm; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NOM - Official Mexican Norm; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TDG - Transportation of Dangerous Goods; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative; WHMIS - Workplace Hazardous Materials Information System

Disclaimer

FMC Corporation believes that the information and recommendations contained herein (including data and statements) are accurate as of the date hereof. You can contact FMC Corporation to insure that this document is the most current available from FMC Corporation. No warranty of fitness for any particular purpose, warranty of merchantability or any other warranty, expressed or implied, is made concerning the information provided herein. The information provided herein relates only to the specified product designated and may not be applicable where such product is used in combination with any other materials or in any process. The user is responsible for determining whether the product is fit for a particular purpose and suitable for the user's conditions and methods of use. Since the conditions and methods of use are beyond the control of FMC Corporation, FMC Corporation expressly disclaims any and all liability as to any results obtained or arising from any use of the products or reliance on such information.

BR / EN